

Sequences, Progressions & Recurrences

IOQM Grade 9 - Assimilation + First-Step Pack

For the learner who remembers AP/GP formulas but does not yet reliably recognize the engine of a mixed sequence problem.

Validated anchors: IOQM-2025-Q26; IOQM-2023-Q10. Exact historical wording remains under source-paper custody.

Concept map

Governing idea: compare neighboring indices so most of the structure cancels. $S_n - S_{(n-1)}$ isolates a term; adjacent moving windows isolate the entering and leaving terms; shifted recurrences often reveal a simpler difference sequence.

Reconnect

3,7,11,15,... $\rightarrow$ constant difference.

2,6,18,54,... $\rightarrow$ constant ratio.

Partial sums S_n given, term a_n asked $\rightarrow$ subtract adjacent partial sums.

Moving window increases $\rightarrow$ subtract adjacent windows.

Term vs sum

If $S_n = a_1 + \dots + a_n$, then $a_n = S_n - S_{(n-1)}$. This converts accumulated information into one local contribution.

Window cancellation

For $W_i = a_i + \dots + a_{(i+k-1)}$, $W_{(i+1)} - W_i = a_{(i+k)} - a_i$. Thus increasing 4-term averages imply $a_{(i+4)} > a_i$; decreasing 7-term averages imply $a_{(i+7)} < a_i$. This is the mechanism behind IOQM-2025-Q26.

Recurrence transformation

If $a_{(n+2)} = 3a_{(n+1)} - 2a_n$, then $a_{(n+2)} - a_{(n+1)} = 2[a_{(n+1)} - a_n]$. The first differences form a GP. A high-index recurrence is often a transform problem, not an invitation to list terms.

Telescoping

$1/[k(k+1)] = 1/k - 1/(k+1)$. Internal terms cancel; boundary terms survive.

First-Step Reference

partial sum given, term asked $\rightarrow$ subtract adjacent sums

constant difference $\rightarrow$ AP

constant ratio $\rightarrow$ GP

moving sums/averages $\rightarrow$ subtract adjacent windows

high-index recurrence $\rightarrow$ transform before brute force

consecutive denominator factors $\rightarrow$ partial fractions / telescope

H0 mastery

1) $S_n = 4n^2 + n$: find a_n .

2) $a_{(n+2)} = 5a_{(n+1)} - 4a_n$, $a_1 = 2, a_2 = 5$: find a_8 efficiently.

3) Evaluate sum from $k=1$ to 50 of $1/[k(k+1)]$.

4) Every 6-term sum is larger than the preceding one: derive the termwise inequality.

5) Equal adjacent 4-term sums: prove period dividing 4.

6) $a_1 = 3, a_2 = 7$, $a_{(n+2)} = 3a_{(n+1)} - 2a_n$: find a_{20} without listing 20 terms.

Adopt: compare neighboring indices so cancellation exposes the mechanism.